# FahMai RAG

## SuperAI Level 1 Hackathon— 2026

Build AI that actually understands Thai customer questions.

10th ANIVERSARY

# Fun Fact

Season 5 has no actual RAG hackathon!

# The Story

FahMai (ฟ้าใหม่) is a fictional Thailand's electronics store.

## 5 House Brands

SaiFah · DaoNuea · KluenSiang
WongKhoJon · JudChuem

## 110 Products

Phones, Laptops, Audio, Wearables, Accessories

## 5 Branches

Siam · Lad Phrao · Rama 9
Chiang Mai · Phuket

# The FahMai Universe

### SaiFah
Phones & Tablets

23 products

### DaoNuea
Computers & Laptops

23 products

### KluenSiang
Audio

22 products

### WongKhoJon
Wearables

12 products

### JudChuem
Accessories

18 products

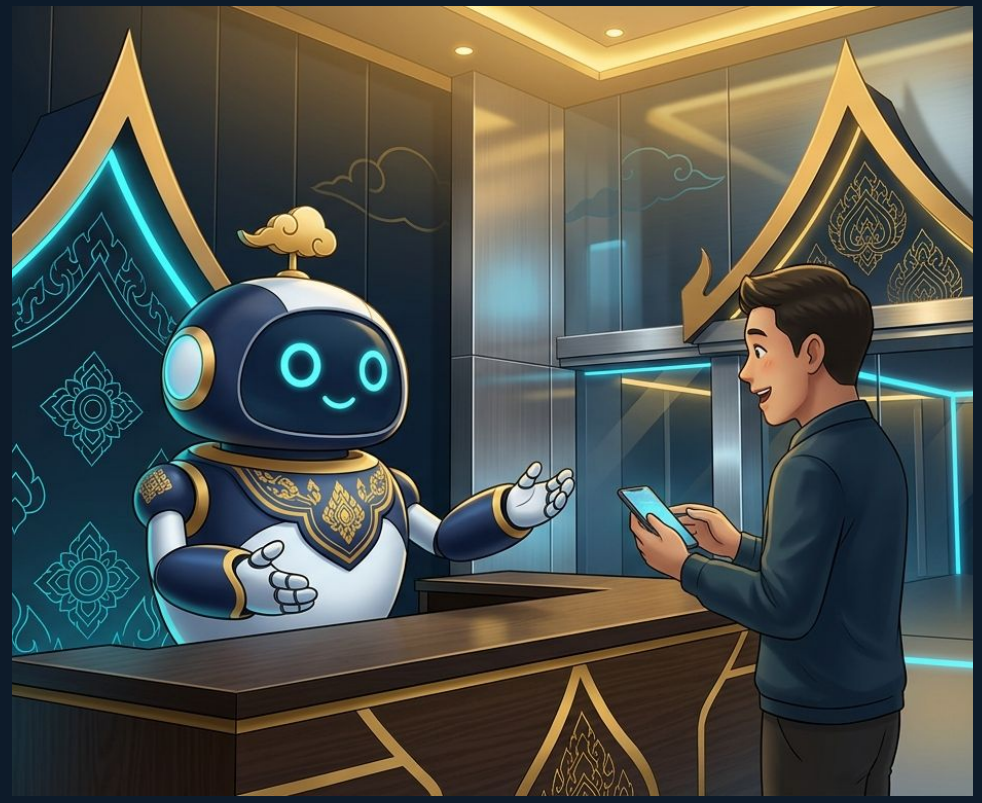
**Partner Brands**: ZenByte (Storage) · NovaTech (Peripherals) · PulseGear (Power) · ArcWave (Displays)

# Your Mission

**Congratulations! You've just been hired as FahMai's first AI Customer Service Agent.**

Customers send questions in casual Thai about products, warranties, returns, shipping, and membership points.

**Your job**: build a RAG system that reads the knowledge base and answers their questions correctly.



# How RAG Works

### 1. CHUNK

Split 135 docs into small passages

### 2. EMBED

Convert chunks to vector representations

### 3. RETRIEVE

Find most relevant chunks for a question

### 4. GENERATE

LLM reads chunks and picks the answer

**Knowledge Base:** 127 products + 5 policies + 3 store info files = 135 markdown documents
The starter kit notebook walks you through Dense, BM25, and Hybrid (RRF) retrieval.

# Competition Format

## Question Format

100 multiple-choice questions in Thai

10 choices per question

Choices 1-8: content answers

Choice 9: "ไม่มีข้อมูลนี้ในฐานข้อมูล" (Not in KB)

Choice 10: "คำถามนี้ไม่เกี่ยวข้อง" (Not related)

## Submission

CSV file: id, answer

answer = integer 1-10

Public leaderboard: 60% of questions

Private leaderboard: 40% (final rank!)

Metric: Accuracy

Higher = Better

**Example Question:**

**"Watch S3 Ultra กันน้ำได้กี่ ATM ครับ?"**

1) 3 ATM 2) 5 ATM 3) 10 ATM 4) 20 ATM 5) 50 ATM ... 9) ไม่มีข้อมูล 10) ไม่เกี่ยวข้อง

# Your Starter Kit

### Knowledge Base

118 markdown files
(products, policies, store info)

### questions.csv

100 questions
10 choices each, no answers

### Starterkit Notebook

Dense, BM25, Hybrid
walkthrough + Typhoon LLM

### ThaiLLM API Key

Access to Typhoon,
OpenThaiGPT, Pathumma, KBTG

# Ideas to Explore

### Better Embeddings

Try stronger multilingual embedding

### Try Other ThaiLLMs

**Must use** ThaiLLM: typhoon, openthaigpt, kbtg, pathumma

### Chunk Size Tuning

Split by structure or other methods or add useful information to each chunk

### Hybrid Retrieval

Dense + BM25 via RRF
→ keywords + semantics

### Prompt Engineering

Adjust the system prompt, add few-shot examples, or change the output format

### Reranking

Cross-encoder or LLM-based
→ better top-K selection

# Let's Hack!

**Starter kit notebook:** https://colab.research.google.com/drive/1GMWk8NuUh1SCv3kK7zLEdns9avc9qnb3?usp=sharing

**Kaggle link**: https://www.kaggle.com/t/e5771bca3ab441e4a1e101839b9098ec

**Deadline**: Sunday 29 @ 8:00 am

**Must use**: ThaiLLMs

**Importance**: Your submission must be fully automated - no manual answers.

Feel free to implement your own RAG!!